

LOGAN-CACHE, UT, USA

WMO: 724796

Lat: 41.787N

Lon: 111.853W

Elev: 1358

StdP: 86.04

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94128

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.2	-22.5	0.6	-20.3	-19.4	0.8	-16.4	9.3	2.9	7.7	1.7	0.5	230	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	34.9	16.4	33.6	16.1	32.2	15.7	18.3	29.6	17.7	29.1	16.9	28.6	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	12.6	21.1	13.7	11.5	20.8	12.6	10.7	20.7	57.0	29.3	54.7	28.8	52.3	28.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.5	5.9	-23.6	37.0	4.1	1.1	-26.6	37.8	-29.0	38.5	-31.3	39.1	-34.3	39.9		
(5)			-23.9	20.2	4.1	2.0	-26.8	21.6	-29.2	22.8	-31.5	23.9	-34.4	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.4	-2.9	3.5	8.0	12.5	17.7	22.7	21.3	15.8	8.7	2.1	-4.2		
	DBStd	10.44	6.29	6.05	4.79	3.66	4.06	3.82	2.28	2.50	3.82	3.95	4.64	5.66		
	HDD10.0	1902	477	360	206	81	22	2	0	0	4	71	237	442		
	HDD18.3	3923	735	593	461	311	184	54	1	6	91	300	486	700		
	CDD10.0	1310	0	0	4	21	99	234	393	350	179	29	1	0		
	CDD18.3	288	0	0	0	0	3	36	136	98	16	0	0	0		
	CDH23.3	5439	0	0	0	9	161	882	2198	1673	494	22	0	0		
	CDH26.7	2550	0	0	0	1	37	362	1172	818	158	3	0	0		
(14) Wind	WSAvg	2.0	1.3	1.6	2.1	2.6	2.4	2.3	2.2	2.2	1.9	1.7	1.6	1.5		
(15) Precipitation	PrecAvg	420	41	34	40	49	52	32	15	21	26	41	33	37		
	PrecMax	638	112	69	92	133	125	96	58	63	83	124	67	106		
	PrecMin	276	9	3	14	11	9	4	0	1	3	0	7	3		
	PrecStd	95	27	17	22	26	32	24	15	16	20	26	16	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	13.6	21.0	24.5	29.9	34.5	37.2	35.6	32.4	26.2	17.7	11.1		
		MCWB	4.1	6.3	9.6	11.7	13.9	16.3	17.4	16.7	15.9	12.2	8.1	5.6		
	2%	DB	6.4	10.5	17.6	21.8	27.4	32.6	35.8	34.0	30.4	22.7	15.1	7.7		
		MCWB	3.1	5.2	8.1	10.0	13.8	15.2	16.6	16.2	14.5	11.2	7.4	4.0		
	5%	DB	4.1	7.6	14.7	19.1	24.9	30.9	34.3	32.8	28.5	20.1	12.4	4.9		
		MCWB	1.8	3.7	6.9	9.1	12.9	15.0	16.4	15.8	13.6	10.3	6.3	2.3		
	10%	DB	2.4	4.9	12.0	17.0	22.4	28.7	32.8	31.4	26.4	17.5	9.6	2.9		
		MCWB	0.8	2.1	5.8	8.4	12.1	14.6	16.2	15.4	13.2	9.2	5.2	1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	6.9	10.3	12.5	16.1	18.2	19.5	18.7	17.7	13.7	9.7	6.4		
		MCDB	8.2	12.2	18.7	23.0	27.0	30.5	29.7	30.0	26.5	22.3	15.3	10.0		
	2%	WB	3.4	5.4	8.8	10.7	14.6	16.8	18.7	17.9	15.9	12.4	8.0	4.3		
		MCDB	5.7	9.8	16.4	19.5	24.7	29.0	29.8	29.5	26.2	19.7	13.4	7.2		
	5%	WB	2.1	3.9	7.4	9.7	13.5	15.8	18.0	17.3	15.1	11.2	6.7	2.7		
		MCDB	4.1	7.2	14.1	17.9	23.4	27.6	29.3	29.0	25.1	18.1	11.7	4.6		
	10%	WB	0.8	2.4	6.0	8.7	12.4	15.1	17.4	16.5	14.1	9.9	5.4	1.3		
		MCDB	2.1	4.8	11.1	15.9	21.3	26.6	28.8	28.3	23.9	17.0	9.5	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.7	11.6	13.2	14.8	18.0	19.7	19.4	18.3	15.5	11.9	8.6		
		MCDBR	9.6	11.4	17.0	18.7	20.1	22.0	22.2	21.8	22.0	20.3	16.3	10.2		
		MCWBR	6.7	7.0	8.8	9.0	8.6	7.8	7.2	7.2	8.3	10.0	9.4	7.0		
		MCWBR	9.0	10.8	16.2	16.9	18.4	19.7	18.8	18.5	18.3	17.7	15.1	9.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.254	0.268	0.287	0.316	0.330	0.320	0.349	0.349	0.318	0.295	0.267	0.253		
		taud	2.405	2.422	2.492	2.430	2.440	2.507	2.444	2.427	2.503	2.546	2.558	2.482		
		Ebn at Noon	912	952	969	955	944	950	918	909	921	909	896	888		
		Edh at Noon	77	90	95	110	112	105	111	109	94	79	67	66		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.90	2.76	4.18	5.42	6.41	7.69	7.58	6.56	5.24	3.54	2.23	1.54		
		RadStd	0.21	0.32	0.29	0.34	0.57	0.55	0.34	0.30	0.35	0.24	0.23	0.20		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page